Software Testing and Validation
A.Y. 2026/2027
Exam Project Specifications

Igor Melatti

1 How to Submit the Project

In this document, we describe the exam project specifications. You must submit a single file `STV_2026_2027_matricola_numbers.zip` (if the project is developed by a group of students, the file name must contain all the students' matricola numbers interleaved by the underscore symbol _). Such a file must contain a single directory `STV_2026_2027_matricola_numbers`, which must include:

- a PDF file `description.pdf` with the following requirements:
  - it must include name, surname and matricola number of each student developing the project;
  - it must describe how the project has been carried out, also pointing out which testing techniques have been chosen and why;
- a PDF file `presentation.pdf`, containing slides to be used to explain the project (assume 20 minutes time for the presentation);
- a directory `project` containing all project files, suitably organized in further subdirectories; such organization must be explained in `description.pdf`.

The resulting `STV_2026_2027_matricola_numbers.zip` file must be sent by email to `igor.melatti@univaq.it`.

It is possible to speak with your colleagues during the project development. However, each group must present a distinct solution.

2 System Under Verification

The System Under Verification (SUV) may be chosen among the following:

- NuSMV, from https://nusmv.fbk.eu/downloads.html;
- SPIN, from https://spinroot.com/spin/Archive/;

3 Specifications for 3-Persons Groups

The project requirements are the following.

1. In the chosen SUV, you must perform
   - unit testing: select 6 units (explain why they are actually units) and test them using the stuiable techniques, from those introduced during the course;
   - integration testing: select 3 sets of units and test them (if needed, you may include some unit not selected in the unit testing phase);
   - system testing;
   - regression testing: also consider the last-but-one SUV version and explain on which parts you may apply the regression testing techniques introduced during the course.
2. Choose one unit and compute MC/DC and loop boundary coverage, by suitably instrumenting the code.

In developing your solution, you are encouraged to use publicly available LLMs (Large Language Models). In such a case, the project relation must show and discuss the submitted prompt and the obtained answers. In particular, you have to discuss if the so-obtained answers are correct/complete, explaining the reason behind that. It is recommended to follow this strategy for the prompt:

1. simply ask the LLM to perform the testing phase as described above;
2. also tell LLM to use the methods we discussed during lessons, possibly describing them;
3. compare the obtained results.

4 Specifications for 2-Persons Groups

As above, with the following restrictions:

- unit testing: select 4 units;
- integration testing: select 2 sets of units;

5 Specifications for 1-Person Groups

As above for the 2-persons groups, with the following furhter restrictions:

- unit testing: select 2 units;
- integration testing: select 1 set of units.